Phase 5: Project Development Phase

Author	Nandini Bade
Program	SkillWallet, ServiceNow Global Certification
Platform	ServiceNow (Service Catalog, Flow Designer, Business Rules)
Document	Project Development Phase - Phase 5: Project Development Phase

Records how the solution was built and tested, with screenshots and step-by-step documentation from the working ServiceNow instance.

In this document

5.1 Development Overview and Build Order
5.2 Environment Setup
5.3 Service Catalog Configuration
5.4 Flow Designer Implementation
5.5 Business Rule: Create Incident If Car Unavailable
5.6 Attach the Flow to the Catalog Item
5.7 Configuration Summary (As Built)
5.8 Testing and Results
5.9 Finalize and Move the Update Set
5.10 Troubleshooting Guide
5.11 Results
5.12 Limitations and Future Enhancements
5.13 Conclusion

5.1 Development Overview and Build Order

Order	What is built	Section
1	Environment and Update Set	5.2
2	Catalog item, variables, and choices	5.3
3	Flow: trigger, approval, condition, task, variables,	5.4

	email	
4	Business rule for unavailable cars	5.5
5	Attach the flow to the catalog item	5.6
6	Test, then finalize and move the Update Set	5.8, 5.9

5.2 Environment Setup

• A ServiceNow developer instance with administrative access.
• Service Catalog and Flow Designer (Workflow Studio) available.
• A named Update Set selected as the current set before starting work.

5.3 Service Catalog Configuration

Activity 1: Create the catalog item

1. Open ServiceNow, click All, and search for Maintain Items.
2. Click New and fill in the details: name Car Rental Request, catalog Service Catalog, category Transportation Services, short description Request a car for rental.
3. Click Save. The item is Active and its State is Published.

Activity 2: Create the catalog variables

After saving, open the Variables related list and create the seven variables:

1. Car Type (Select Box)
2. Requested Date (Date)
3. Pickup Location (Single Line Text)
4. Duration Hours (Numeric Scale)
5. Requestor (Reference to sys_user)
6. Drop Location (Single Line Text)
7. Reason (Multi Line Text)

For the Car Type variable, open Question Choices and add three choices: Mini, SUV, and Sedan.

5.4 Flow Designer Implementation

Activity 1: Create the flow and trigger

1. Open All, search for Flow Designer, click New > Flow.
2. Name: Car Rental Req. Run context: System User.
3. Trigger: Service Catalog.

Activity 2: Ask For Approval

1. Add action Ask For Approval.
2. Record: Requested Item Record (sc_req_item).
3. Approval Field: Approval. Journal Field: Approval history.
4. Rules: Approve when Anyone approves, select designated approver.

> **CONCEPT: Ask For Approval**
> The action sets the approval field on the RITM to Requested, creates approval records for the approver, and pauses the flow. When the approver decides, the field is updated and the flow resumes.

Activity 3: Add the condition

1. Add If flow logic step with label If Approve.
2. Condition: Ask For Approval > Approval State IS Approved.

Activity 4: Create Catalog Task

1. Add action Create Catalog Task on table Catalog Task [sc_task].
2. Fields: Assignment group = Transport Team, Description = 'Verify This Item And Assign To A Person Who Manages'.
3. Template catalog item: Car Rental Request. Wait option = enabled.

> **CONCEPT: Wait Option**
> With Wait enabled, the flow pauses until the catalog task is closed by the Transport Team before proceeding to send the completion email.

Activity 5: Get Catalog Variables

Add action Get Catalog Variables, set Submitted Request to trigger Requested Item, and select requestor, requested_date, pickup_location, drop_location, duration_hrs, car_type.

Activity 6: Send Email

Add action Send Email. Target Record: Requested Item (sc_req_item). To: Get Catalog Variables > requestor > Email. Subject: 'Your Car Request is approved.'. Body: 'Dear, Your car Rental Request is approved! Regards.' Save and Activate the flow.

5.5 Business Rule: Create Incident If Car Unavailable

A Business Rule handles the exception case defined on Requested Item [sc_req_item], running after insert and update (Order 100, Advanced).

GlideRecord Script:

```
(function executeRule(current, previous /*null when async*/ ) {
    if (current.state == 3) {
        var inc = new GlideRecord('incident');
        inc.initialize();
```

```
        inc.short_description = 'Car unavailable for request: ' + current.number;
        inc.description = 'Car not available for trip on ' + current.variables.requested_date;
        inc.insert();
    }
})(current, previous);
```

5.6 Attach the Flow to the Catalog Item

Open Car Rental Request catalog item, on Process Engine tab select Car Rental Req flow, save record.

5.7 Configuration Summary (As Built)

Object	Setting	Value
Catalog item	Name / catalog / category	Car Rental Request / Service Catalog / Transportation Services
Catalog item	State / owner	Published / System Administrator (Developer: Nandini Bade)
Variables	Count	7 (Car Type, Requested Date, Pickup Location, Duration (Hours), Requestor, Drop Location, Reason)
Flow	Name / status	Car Rental Req / Active, Published
Flow	Trigger	Service Catalog
Ask For Approval	Field / journal / rule	Approval / Approval history / Approve when Anyone approves
If	Condition	Approval State is Approved
Create Catalog Task	Table / group / Wait	sc_task / Transport Team / enabled
Business Rule	Table / when / order	sc_req_item / after insert and

		update / 100

5.8 Testing and Results

All functional tests (T1-T9) and exception tests (T10-T11) passed successfully on the working ServiceNow instance.

5.9 Finalize and Move the Update Set

1. Mark Update Set state to Complete.
2. Export XML payload and load/commit on target instance.

5.10 Troubleshooting Guide

Symptom	Likely Cause	What to Check
Flow never starts after ordering	Flow is not Active, or trigger is not Service Catalog	Flow status, trigger type, Executions list
No approval appears	Ask For Approval has no valid approver rule	Rules section of Ask For Approval action
Task is not created	Approval State is not Approved	If condition & approval state outcome
Email is not received	Task not completed (Wait enabled) or Requestor has no email	Task state, user email, sys_email log
Incident appears unexpectedly	Business Rule condition matches state 3 unexpectedly	Business rule condition script & state choices

5.11 Results

• Working end-to-end automation from catalog request to approval to fulfillment task to notification.
• Automated exception path creating Incident via Business Rule.
• 100% traceability across REQ, RITM, sc_task, and approval journal entries.

5.12 Limitations and Future Enhancements

• Add dynamic manager approval instead of static rule.
• Implement SLA definitions on RITM and sc_task records.
• Add Catalog Client Scripts for past date validation.

5.13 Conclusion

The project demonstrates how ServiceNow's Service Catalog, Flow Designer, and Business Rules combine to replace manual, email-driven car rental processes with a structured, efficient, and fully auditable workflow.